



Amir Hallaji

ML Engineer / Computer Engineer

Dreams Come True...

Summary

ML Engineer with 4+ years of experience at Snapp (Middle East's largest ride-hailing platform). Combines expertise in backend software engineering with machine learning to develop and deploy production-grade AI systems. Master's graduate in Artificial Intelligence from Shahid Beheshti University, with thesis research in bioinformatics. Proven ability to design, implement, and scale microservices and ML models in high-traffic environments.

Education

- 2022–2025 **M.Sc. in Artificial Intelligence**, *Shahid Beheshti University*, Tehran, Iran
Thesis: Drug-Target Interaction Prediction using Protein and Molecule Language Models
- 2018–2022 **B.Sc. in Computer Engineering**, *Shahid Beheshti University*, Tehran, Iran
GPA: 18.00/20 (Top 10%)
- 2014–2018 **Diploma in Mathematics and Physics**, *Salam Tajrish High School*, Tehran, Iran,
GPA: 19.95/20

Publications

- 2026 **Structure-free drug–target affinity prediction using protein and molecule language models**, *Journal of Cheminformatics*
 - Proposed and implemented a novel deep learning approach using pretrained language models (ChemBERTa and ESM2) to predict drug-target affinity without relying on 3D structural data.
 - Introduced a customized **Residual Inception** neural network architecture to effectively integrate protein and molecule sequence embeddings, achieving multi-scale feature extraction.
 - Achieved **state-of-the-art performance** on benchmark datasets (Davis, KIBA, BindingDB), demonstrating the efficacy of a pure sequence-based method.
 - **Link:** <https://link.springer.com/article/10.1186/s13321-025-01146-6>

Work Experience

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- 2024–Present **ML Engineer, Pricing Team, Snapp Inc., Tehran, Iran**
- Designed and implemented AI pricing models using machine learning and deep learning (PyTorch)
 - Developed data pipeline for Snapp's rider-driver funnel analysis, processing millions of daily events
 - Implemented ClickHouse-based analytics infrastructure for real-time pricing optimization
 - Set up ML infrastructure including model serving, monitoring, and A/B testing frameworks
 - Optimized pricing algorithms resulting in improved matching efficiency and revenue
- 2022–2024 **Software Engineer (Golang), Pricing Team, Snapp Inc., Tehran, Iran**
- Designed and implemented microservices architecture for dynamic pricing system
 - Developed CI/CD pipelines using GitLab, reducing deployment time
 - Managed Kubernetes deployments and implemented auto-scaling solutions
 - Built monitoring dashboards with Grafana and Prometheus for service observability
 - Optimized backend services, improving API response times 3x faster
 - Implemented feature flags and canary deployment strategies
- 2021 **Programming Intern, Backend Development, MavaraTech Company, Tehran, Iran**
- Developed RESTful APIs for e-commerce platform
 - Gained experience with database design and optimization

Technical Skills

Programming Languages	Python, Golang
Machine Learning	Frameworks: PyTorch, Scikit-learn, TensorFlow Domains: Deep Learning, Bioinformatics, NLP
Infrastructure & DevOps	Containerization: Docker, Kubernetes CI/CD: GitLab CI/CD Monitoring: Grafana, Prometheus
Databases	Analytical: ClickHouse Relational: MySQL, PostgreSQL In-memory: Redis
Data Engineering	Data Pipeline Design, Real-time Analytics, Batch Processing
Tools	Git, Linux, Jupyter, MLFlow, REST APIs, Microservices Architecture, Prometheus, Grafana, Jaeger, K8s, Helm

Projects

Snapp AI Pricing System	Development of ML-powered dynamic pricing system using deep learning models. Implemented real-time inference pipeline serving 1k+ requests per second.
ClickHouse Analytics Platform	Designed and deployed ClickHouse cluster for pricing analytics, reducing query latency from minutes to sub-second response times.
Drug-Target Interaction Prediction	Master's thesis: Developed novel deep learning architecture for predicting drug-target interactions using transformers and attention mechanisms.
Microservices Architecture	Built scalable microservices for pricing team, implementing circuit breakers, rate limiting, and distributed tracing.

Teaching Experience

2019–2023 **Programming Instructor**, Salam Tajrish High School, Tehran, Iran
Teaching Introduction to Programming to high school students

2019–2025 **Teaching Assistant**, Shahid Beheshti University

- Machine Learning
- Advanced Programming
- Discrete Mathematics
- Technical English for Computer Engineering

Honors & Awards

2023 **Irancell Labs Winner**
Ranked 1st in AI Hackathon

2017 **Iran Language Institute**
Completed Advanced English proficiency levels

Languages

Persian Native Speaker

English Professional Proficiency (Reading/Writing/Speaking)

Interests

Research Machine Learning, Deep Learning, LLM, NLP, Bioinformatics, Computational Biology
Technology Distributed Systems, MLOps, Real-time Analytics